Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Materials Science and Engineering

MSE 4101
(Modern Iron and Steel Making)

Time: 3 hours Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Describe how the fusion zone forms inside an iron blast furnace and how the formation of this zone affects the operation of blast furnaces. (16)

   (b) Explain how and where impurities (Si, Mn, P and S) are picked up in liquid iron in a blast furnace. (12)

   (c) Is phosphorus reduction possible in a blast furnace? Justify your answer. (07)

2. (a) Explain the properties of coke required for smooth and efficient operation of an iron blast furnace along with their effects on the productivity. (20)

   (b) Explain the electro-chemical nature of slag metal reaction as metal is non polar and molten slag is ionic in character. (10)

   (c) Why open-top ladles were gradually discarded? Explain in brief. (05)

3. (a) Mini blast furnace and electro thermal process are two established processes of producing liquid pig iron in small quantities. Out of these two processes select the most suitable one for Bangladesh and justify your selection. (18)

   (b) Compare and contrast HYL III and HYL IV process. Out of these two processes select the most suitable one for Bangladesh and justify your selection. (17)

4. (a) With proper reaction, describe how carbon bearing ferromanganese is produced. (10)

   (b) A blast furnace produces pig iron of composition of Fe-93.0%, C-3.6%, Si-2.0%, Mn-0.5%, P-0.9%. The flux amounts to one-fourth the weight of the ore. Consumption of coke is 780 kg/ton of pig iron. The gases carry 2 parts CO to 1 part CO_2. Assume that 99.0% of the iron is reduced and 1% iron is slagged. Calculate (25)

   (i) the weight of the ore required to produce one ton (1000kg) pig iron.

   (ii) the weight of slag made from one ton pig iron and its composition

   (iii) volume and composition of the blast furnace gas.

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Hot metal of composition 0.8% Si, 0.2% P, 0.25% Mn, 4% C and in house steel scarp is refined in a converter to produce steel of composition 0.2% C and rest iron. During refining scarp is charged whose amount is 20% of hot metal. Pure oxygen is blown. The composition of slag is CaO 54%, FeO 18% and MnO 2.5 % with CaO/SiO_2=3.5. Exit gases analyses 20% CO_2 and 80% CO. Calculate amount of steel, slag, oxygen and waste gases per ton hot metal. (20)

   (b) Why production of stainless steel is so important for our daily life? (10)

   (c) If the liquid steel is not deoxidized properly, how will it affect in service? Explain in brief. (05)

6. (a) Using schematic diagram, discuss how carbon, silicon, manganese, phosphorous and sulphur are removed in the BOS process. (20)

   (b) "Look after your slag, the steel will look after itself". Do you agree or not? Explain your reason. (10)

   (c) What de-sulphurization is essential for steel making? (05)

7. (a) With schematic diagram, explain how oxygen transfer from gaseous phase through slag into metal. (15)

   (b) Analyze the potential of using pig iron and sponge iron as an alternative charge material to steel scarp in EAF steel making. (12)

   (c) EAF is better that induction furnace-justify. (08)

8. (a) What is secondary steel making? Indicate its principal function. (15)

   (b) The initial temperature of the billet was 1000^0C and rolling time for the first pass was 5 seconds. For subsequent passes, the rolling time was proportional to the length of the product. Find out the final temperature of the rolled steel bar. The average specific heat of steel at high temperature is 0.166 kcal/kg^{0}C. (10)

   (c) Special and alloy steels are almost exclusively produced by arc furnace. Explain the quotation. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Materials Science and Engineering

MSE 4103

(Engineering Alloys and Materials Selection)

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.
(iii) Assume reasonable data if any missing.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) What are the carbide phases present in stainless steel? What role do these phases play regarding the mechanical properties of stainless steel? (15)

   (b) How does distortion develop in tool steels? How can distortion be reduced? (10)

   (c) Draw a heat treatment cycle of 25% Ni maraging steel and explain the heat treatment process. (10)

2. (a) How can you obtain the highest dimensional stability in tool steels? Explain in brief. (15)

   (b) How does Zn affect the properties of Cu-Sn alloy and Sn affect the properties of Cu-Zn alloy? (12)

   (c) What is Muntz metal? Why is it heat-treatable? (08)

3. (a) Draw the phase diagram of Copper-Tin system and discuss the properties of Copper-Tin alloy. (15)

   (b) Describe the properties and heat treatment process of Beryllium bronze. (12)

   (c) What are the functions of grain refiners and modifiers in Aluminium alloys? (08)

4. (a) What is Hydrogen embrittlement? How does Hydrogen embrittlement affect the properties of Titanium alloys? (10)

   (b) Which of the nickel-based alloys would you recommend for nuclear reactors? How does this alloy attain the properties suitable for aforementioned application? (10)

   (c) How does NiTiNOL recover previously defined shapes after being subjected to thermomechanical cycle? Give some typical applications of NiTiNOL. (15)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) What is design? If you are going to design something, what sort of materials information do you need? (05)

   (b) A panel is a flat slab, like a table top. Its length L and width b are specified but its thickness h is free, as shown in figure 5(b). (10)

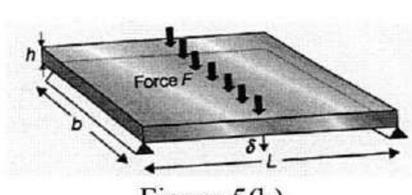


Figure 5(b)

It is loaded in bending by a central load F. The stiffness constant S^* requires that it must not defect more than δ. The objective is to achieve this with minimum mass, m. Derive material index for that panel.

(c) When a pressure vessel has to be mobile, its weight becomes important. Aircraft bodies, rocket casings and liquid natural gas containers are examples; they must be light and at the same time they must be safe. That means they must not fail by yielding or by fast fracture. (20)

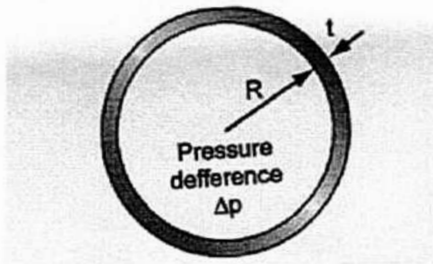


Figure 5(c)(i)

Assume for simplicity that it is spherical of specified radius R and that the wall thickness t is small compared with R. Then the tensile stress in the wall is $\sigma = \frac{\Delta PR}{2t}$.

(i) Derive Material Indices M_1 and M_2.

(ii) Make a coupling relation between them.

(iii) Determine the gradient and intercept of this line and plot it for first c=5 mm and then c=5μm in figure: 5(c)(ii). Identify the lightest candidate materials for the vessel for each case. M_1 and M_2 need to be minimized to find the lightest material [Attach figure 5(c)(ii) with the answer].

6. (a) Explain the concept of exchange constant. (10)

(b) The exchange constant for weight saving in light trucks is $\alpha = \$12/kg$, meaning that the value of weight reduction over the life of the vehicle is $12 for each kilogram saved. A maker of such vehicles offers three models. The first uses steel panels for the body work. The second uses aluminium, costs $2,500 more but weighs 300 kg less. The third offers carbon-fiber paneling, costs $8,000 more, and weighs 500 kg less. Which is the best buy? (10)

(c) Aircraft and space structures make use of plates and shells. The index depend on the configuration. Here you are asked to derive the material index for a circular plate of radius a, carrying a centered load W with a prescribed stiffness $S = \frac{W}{\delta}$ and of minimum mass, as shown in the figure: 6(c). (15)

Use the two results listed below for the mid-point deflection δ of a plate or spherical shell under a load W applied over a small centered, circular is in which $A \approx 0.35$ is a constant. Here, E is Young's modulus, t is the thickness of the plate or shell and ν is Poisson's ratio. Poisson's ratio is almost the same for all structural materials and can be treated as a constant.

Circular plate: $\delta = \frac{3}{4\pi}\frac{Wa^2}{Et^3}(1-\nu^2)(\frac{3+\nu}{1+\nu})$

Hemispherical Shell: $\delta = A\frac{Wa}{Et^2}(1-\nu^2)$

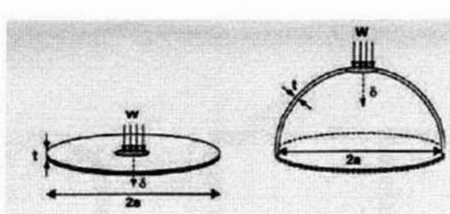


Figure 6(c)

7. (a) Explain the concept of penalty function. (10)

(b) Calculate the gain in bending efficiency Ψ_B^e when a solid is formed into small, thin-walled tubes of radius r and wall thickness t that are then assembled and bonded into a large array, part of which is shown in the figure 7(b). Let the solid of which the tubes are made have modulus E_s and density ρ_s. Express the result in terms of r and t. (10)

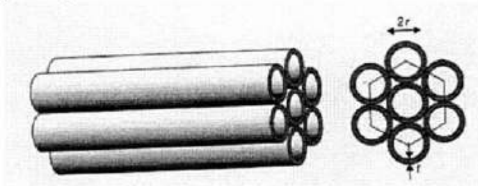


Figure 7(b)

(c) Derive the expression for the shape-efficiency factor Φ_B^e for stiffness-limited design for a beam loaded in bending with each of the three sections listed below. Do not assume that the thin-walled approximation is valid. (15)

(i) a closed circular tube of outer radius $5t$ and wall thickness t,

(ii) a channel section of thickness t, overall flange width $5t$ and overall depth $10t$, bent about its major axis; and

(iii) a box section of wall thickness t and height and width h_1=$10t$.

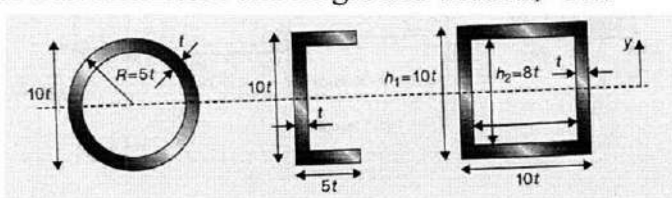


Figure 7(c)

8. (a) Why percolation is an important design tool for hybrids? (05)

(b) Explain compressive deformation behavior of foams with neat sketches. (10)

(c) Given figure: 8(c) is a chart for exploring stiff composites with light alloy or polymer matrices. The criteria of excellence (the indices E/ρ, $E^{1/2}/\rho$ and $E^{1/3}/\rho$) for light and stiff structures are shown; they increase in value towards the top left. (20)

(i) Use the chart to compare the performance of a titanium-matrix composite reinforced with zirconium carbide (ZrC), saffil alumina fibers and Nicalon silicon carbide fibers.

(ii) Use the chart to explore the relative potential of Magnesium-E glass-fiber composites and Magnesium-Beryllium composites for light, stiff structures.

Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Materials Science and Engineering

MSE 4105
(Polymer and Composite Materials)

Time: 3 hours

Full Marks: 210

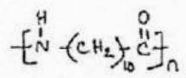
N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

Q 1. (a) Briefly explain the main steps of polymerization with proper examples. (10)

(b) Nylon 11 has the following structure (10)



If the number average degree of polymerization, $\overline{DP}_n$ for nylon 11 is 100 and $\overline{M}_w = 120{,}000$. What is its polydispersity?

(c) Derive the kinetics for both noncatalysed and acid catalyzed step growth polymerizations. Which poly-condensation polymerization reacts faster? Explain from the typical *DP* versus *t* plot. (15)

Q 2. (a) What are the polymer additives and why these additives are used in polymer processing? (10)

(b) What are the different polymer processing techniques? Briefly explain the polymer fiber spinning process with neat sketch. (15)

(c) What is Kevlar polymer? Write down its properties, synthesis reaction and applications. (10)

Q 3. (a) What is meant by degree of crystallinity of polymer? Explain the factors affecting the melting and glass transition temperature of polymers. (10)

(b) Discuss the stress-strain behaviors of different polymeric materials with proper diagrams. (07)

(c) Suppose you have two laminated composites, one is fabricated by polyester resin and wood fiber another in made by polypropylene and wood fiber. Polyester and polypropylene have elastic modulus of 6×10^5 psi and 2×10^5 psi respectively and Poisson's ratio of 0.37 and 0.30 respectively. For a strain of 0.05 (18)

(i) Calculate the shear stress and the percentage change in volume of both polymers.
(ii) On the basis of your result from (i), find out which composite is more durable in shear force loading and why?
(iii) If you have to improve the efficiency of that less durable composite, then what parameters you have to correct or add to the fabrication process.

Q 4. (a) Define co-polymerization. Explain the formation of nano-rod structure of metal using PEO-b-PMA block copolymer template. (15)

(b) What is self-assembled monolayers? Briefly discuss the formation of SAMs on a substrate with proper example and figures. (10)

(c) Write short notes on (10)
(i) Liquid crystal polymer
(ii) Conducting polymer.

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

Q 5. (a) Define composite materials. How composite materials are formed? Explain with necessary figures. (12)

(b) How composite materials can bear higher load than conventional materials? Explain with necessary figures. (12)

(c) Why random fiber composite exhibit same mechanical properties at different orientation than Uni and Bi-directional fiber composite. Explain using figure 5(c). (11)

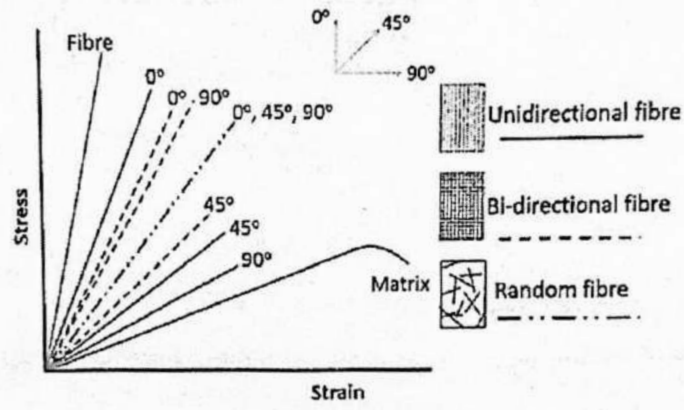


Figure 5(c)

Q 6. (a) Define fiber. Write a short note on glass and metallic fibers. (12)

(b) Describe different defects of composite materials with proper explanations. (10)

(c) Write a short note on Filler and Additives. (07)

(d) Define Whiskers. Why whiskers are very expensive? (06)

Q 7. (a) Write a short note on any two thermosetting resin from below (10)

(i) Epoxy
(ii) Polyester
(iii)Vinyl ester.

(b) Describe a synthesis technique for metal matrix composite with appropriate figures. (08)

(c) Write a short note on ceramic matrix composite with advantages, disadvantages and applications. (11)

(d) What are the factors that affecting the properties of polymer matrix composite. (06)

Q 8. (a) What is work of adhesion? Show that when contact angle, $\theta = 0°$ the work of adhesion is $W_a = 2\gamma_{LV}$ and when, $\theta = 180°$ the work of adhesion is $W_a = 0$. (14)

(b) Write down the equation for modulus of composite in transverse direction. (12)

(c) Write a short note of hybrid composite. (09)

Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Materials Science and Engineering

MSE 4107
(Computational Materials Science)

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Define computational materials science (CMS) and Computer Materials Engineering (CME). Describe how CMS and CME are different. (13)
   (b) How modeling, simulation and analysis are related? Explain with proper flow chart. (13)
   (c) When computer simulation is not appropriate tool? Explain in brief. (09)

2. (a) Write down the goals of computational materials science. Explain the basic procedures of computational materials science. (13)
   (b) How does computer simulations data support the experimental results? (10)
   (c) Write down some state equations with corresponding state quantity and state variable(s) in computational materials science. (12)

3. (a) Define density functional theory (DFT). Write down the applications of density functional theory. (12)
   (b) Write down Hohenberg-Kohn theorems with necessary figures. Proof that, the external potential is a unique functional of the density. (16)
   (c) Explain the challenges of Schrodinger equation solution for many body system. (07)

4. (a) Describe Rietveld refinement process with proper figure. Explain how materials scientist become benefited from this process. (13)
   (b) Describe self-consistent procedure to solve Khon-Sham equations with appropriate explanation and flow charts. (14)
   (c) Write a short note on Hamiltonian operator. (08)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) What is the role of potential in a Molecular Dynamics simulation? Draw the schematic representation of Lennard-Jones potential and explain its several features. (15)
   (b) Express the Lennard-Jones pair potential with parameters and determine the Lennard-Jones force. (10)
   (c) Describe pros and cons of the Tersoff potential. (10)

6. (a) How can you solve Newton's equations of motion for a set of atoms to obtain material properties in Molecular Dynamics simulation? (15)
   (b) How does NVE ensemble differ from NVT ensemble? (08)

(c) Au nanowire is made of 8000 atoms in an FCC structure. The aim is to identify the yield mechanism of the Au nanowire under uniaxial tensile deformation. Which potential will be best suited for this MD simulation? Explain your selection. (12)

7. (a) How does Verlet algorithm work? Also, mention its advantages and disadvantages. (12)

(b) Consider a one-dimensional harmonic oscillator of mass, m, that follows Hooke's law, $F = -kx$, in an isolated system (E=const). Then Newton's second law becomes the second-order ordinary differential equation, (16)

$$a(t) = -\frac{k}{m}x(\mathrm{t})$$

Let us assume that the spring is fixed at one side and is an ideal spring with zero mass and spring constant k. Given the oscillator frequency $\omega = \sqrt{\frac{k}{m}} = 1$ and initial conditions $x_o = 5$ and $v_o = 0$, find and discuss the followings:

(i) Write down the potential expression for the system.

(ii) Numerical solution for ball positions with time by taking the first two terms in the Taylor series expansions (the Euler methods) with $\Delta t = 0.2$ for 5 timesteps.

(iii) Discuss what will happen if the numerical calculation proceeds further for more timesteps and how to improve its accuracy?

(c) Explain how dynamical properties are calculated from Molecular Dynamics simulation. (07)

8. (a) Explain the principle of Monte Carlo method. Also, mention its limitations. (13)

(b) How does time average different from ensemble average? (07)

(c) Consider the simple two-state system in which energy in state 1 is E_1 and in state 2 is E_2. Assume $E_2 > E_1$, the $\Delta E = E_2 - E_1$, and $\Delta E / K_B T = 1$, with the starting state as state 1. Generate a series of random number and explain how the Metropolis Algorithm works. (15)

Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Materials Science and Engineering

MSE 4109
(Electronic, Magnetic and Optical Properties of Materials)

Time: 3 hours Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

Q 1. (a) Suppose you have to select a copper alloy for the windings of an electromagnet designed to create very high magnetic field. The windings must sustain stresses much higher than pure Cu can handle. Explain what kind of alloying elements you will use to increase in strength with minimum decrease in conductivity. (10)

(b) Because of a shortage of Cu during World War II, the electromagnets used in Oak Ridge to separate U-235 isotopes for the Hiroshima atomic bomb were wound with pure silver, which has an atomic volume of 10.30 cc/mol. If 100 m of silver wire 2 mm in diameter required 8.1 watts electrical power when carrying 4 amps of current, calculate the (i) resistivity; (ii) electron mobility; (iii) collision time; (iv) mean free path; (v) Hall coefficient and (vi) Hall field in a transverse magnetic field of 9 tesla. You may remember that electrical power equals the product of current & voltage. Assume T = 300 K, silver is monovalent (one free electron per atom) and any other missing data. (18)

(c) Why does the resistivity of Cu increase with increasing temperature? (07)

Q 2. (a) Derive an equation that relates microscopic electronic polarization(α_e) to the macroscopic relative permittivity(ε_r). (12)

(b) Diamond has a dielectric constant of 5.68. Calculate the polarization, electric displacement and dielectric susceptibility when diamond is exposed to an electric field of 1V/mm. (08)

(c) What is the Heisenberg uncertainty principle? Explain the Heisenberg uncertainty principle applying the Schrodinger's wave equation, consider the motion of an electron in free space. (15)

Q 3. (a) Estimate the probability of an electron with energy E = 0.12 eV tunneling through a rectangular potential barrier with a height of V_o = 1.2 eV and a width of 5Å. (08)

(b) Prove that all extrinsic semiconductors either p-type or n-type transform into intrinsic semiconductors at high temperature. (10)

(c) Demonstrate that the first derivative of E with respect to K is related to the velocity and the second derivative is related to the mass, of a free electron. Use these results, explain the concept of effective mass and prove that the effective mass of an electron near the top of the valence band in negative. (17)

Q 4. (a) Derive an expression for the density of states for a 1D quantum wire. (15)

(b) Determine the number of quantum states per cm^3 in silicon between E_c and $(E_c + 0.32)$ eV at T=300K using the basic principle of infinite potential well in 3D. (10)

(c) Draw the E versus k diagram of the conduction and valence bands of a semiconductor at (a) T= 0 K and (b) T > 0 K in the absence of any external force and prove that net drift current is zero. (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

Q 5. (a) Schematically show the E vs K diagram of GaAs and Si and explain the disadvantage of using Si for use in semiconductors lasers and other optical device. (GaAs direct band Gap material, Si indirect band Gap material) (15)

(b) "Radiation is amplified in LASER". Justify the statement with diagram. (10)

(c) Explain absorption process. Describe the absorption of light by inter-band and intra-band transitions. (10)

Q 6. (a) 'A good conductor such as Cu is a good reflector in the IR region'. Prove it with mathematical expression. (15)

(b) Describe the courses of dispersion of light. (10)

(c) At the frequencies of visible light, a transparent plastic has an effective dielectric constant ε_r of 1.7. If surrounded only by air, what will be the critical angle for internal reflection? What will be the reflectivity for normal incidence? Explain the photoconductivity phenomenon. (10)

Q 7. (a) "A material prefers to be magnetized along an easy axis". Discuss this statement with the help of anisotropy energy, anisotropy field and domain wall surface. (10)

(b) An H field of 50A/M is applied to a ferromagnet and total magnetic field B within the sample measured to be 2 Tesla. If the saturation magnetization is 1.7×10^6 A/m. What percentage of the sample consist of domains magnetized parallel to the field? What is the value of magnetic susceptibility? (10)

(c) With sketch, describe idealized magnetization curve of a reversible ferromagnetic material with domain wall. (15)

Q 8. (a) Describe with illustration, the BCS theory of superconductivity in terms of electron-phonon interaction. (17)

(b) Diamagnetic susceptibility is negative and independent of temperature. Explain it with mathematical expression. (18)